HRFuse-Net: Hierarchical Resolution-Retentive Feature Encoding for Brain Metastasis Segmentation

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ABSTRACT

Accurate segmentation of small brain metastases in 3D MRI is a critical yet challenging task, as lesions occupying only a few voxels are highly susceptible to the loss of fine spatial detail in standard encoder-decoder architectures. To address this, we propose HRFuse-Net, a novel segmentation framework built on the principle of hierarchical resolution-retentive feature encoding. The architecture introduces two core components: a Parallel High-Resolution Residual Stream (PHRS) that maintains a dedicated feature pathway at the original input resolution throughout the encoder, and an Axial Strip Context Aggregator (ASCA) that efficiently captures long-range, directional contextual information. The PHRS prevents the irreversible compression of sub-voxel boundary cues by fusing high-resolution details into each encoder stage via a learned gated mechanism, while the ASCA enriches semantically rich features with axis-wise context using a computationally efficient strip-pooling strategy. Validated on the publicly available LMD-BM dataset, HRFuse-Net demonstrates superior performance in delineating the necrotic core, peritumoral edema, and enhancing tumor subregions, particularly for small metastases, establishing a new state-of-the-art for this clinically vital task. The code and checkpoints are available at https://github.com/yifangao112/HRFuse-Net

1. Introduction

Accurate segmentation of brain metastases from magnetic resonance imaging (MRI) is a cornerstone of modern neuro-oncology, enabling precise treatment planning, response assessment, and prognostication Wang, Hsu, Lee, Pan, che Yang, chia Lee, and Wu (2023). While deep learning, particularly encoder-decoder architectures like U-Net Ronneberger, Fischer, and Brox (2015) and its variants, has achieved remarkable success in segmenting large anatomical structures, the reliable delineation of very small lesions remains a formidable challenge. This is especially critical for brain metastases, where early detection and accurate volumetric measurement of sub-centimeter lesions are essential for guiding stereotactic radiosurgery and monitoring disease progression Kanakarajan, Baene, Hanssens, and Sitskoorn (2025).

The core difficulty lies in the fundamental conflict between spatial detail preservation and contextual feature extraction within standard segmentation networks. For lesions smaller than 0.02 cc, which may occupy only a handful of voxels in a 3D MRI volume, accurate boundary delineation is critically dependent on preserving fine-grained spatial information throughout the network's processing depth. Conventional encoder-decoder architectures apply successive downsampling operations to expand the receptive field and capture semantic context. However, this process geometrically attenuates spatial resolution at each stage, irreversibly discarding the sub-voxel boundary cues necessary to distinguish tiny, low-contrast metastases from surrounding parenchyma or imaging artifacts Wang, Chen, Wang, and Bao (2024). By the time multi-scale context aggregation modules, such as atrous spatial pyramid pooling (ASPP) or feature pyramid networks Lin, Dollár, Girshick, He, Hariharan, and Belongie (2016), are applied at the deepest encoder stage, the high-resolution information required for precise small-object segmentation has already been lost.

Recent efforts to address this resolution-context trade-off have introduced several promising directions. Transformer-based architectures cheng Liu, Lv, Yang, Li, Lee, and Shen (2022) and state-space models use long-range self-attention or selective scan mechanisms to capture global context without aggressive downsampling, but often incur prohibitive computational costs for high-resolution 3D data. Techniques employing atrous convolutions uniformly across stages Chen, Papandreou, Schroff, and Adam (2017) can

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expand receptive fields but fail to recover the high-frequency detail lost in earlier downsampling steps and may introduce grid artifacts in 3D. Self-distillation methods Zhang, Qu, Xie, Li, Zheng, and Li (2021) attempt to transfer fine-grained knowledge from auxiliary branches, yet they require complex feature alignment and do not fundamentally prevent the initial information loss. Consequently, there remains a significant gap in architectures that can simultaneously maintain a high-fidelity spatial representation and integrate rich, long-range contextual information in a computationally tractable manner for 3D medical image analysis.

To bridge this gap, we propose HRFuse-Net, a novel architecture founded on the principle of hierarchical resolution-retentive feature encoding. Our core insight is to maintain a persistent, parallel stream of features processed at the original input resolution throughout the entire encoder pathway. This Parallel High-Resolution Residual Stream (PHRS) bypasses all downsampling operations, preserving fine spatial detail that is then selectively fused into each encoder stage via a learned gated mechanism before further resolution reduction occurs. We complement this with an Axial Strip Context Aggregator (ASCA), a lightweight module that captures elongated, long-range contextual cues along each spatial axis independently. This design efficiently models directional dependencies crucial for identifying small lesions while avoiding the quadratic memory complexity of full volumetric self-attention.

The principal contributions of this work are as follows. First, we introduce the Parallel High-Resolution Residual Stream (PHRS), a dedicated feature pathway that maintains a fixed high-resolution representation throughout the encoder, enabling the continuous injection of fine spatial detail to counteract the information loss inherent in progressive downsampling. Second, we design the Axial Strip Context Aggregator (ASCA), an efficient module that aggregates long-range contextual information along individual spatial axes via strip pooling and depthwise convolutions, providing essential semantic context for small lesion recognition with linear computational complexity. Third, we integrate these components into a cohesive framework, HRFuse-Net, and demonstrate its efficacy on a publicly available longitudinal brain metastasis MRI dataset. Our method achieves state-of-the-art performance, with a Dice score of 0.5069 and an HD95 of 25.4869, representing a 9.75% improvement in Dice and a 21.67% reduction in HD95 over a strong nnU-Net baseline. Extensive ablation studies confirm the contribution of each proposed component.

2. Related Work

2.1. Encoder-Decoder Architectures for Medical Segmentation

The foundational encoder-decoder architecture for medical image segmentation was established by the fully convolutional network (FCN), which showed that networks composed solely of convolutional layers could produce dense pixel-wise predictions Shelhamer, Long, and Darrell (2014). This paradigm was significantly advanced by the U-Net architecture, which introduced a symmetric encoder-decoder structure with skip connections to combine high-resolution encoder features with upsampled decoder features, enabling precise localization Ronneberger et al. (2015). Extending this to 3D medical volumes, V-Net incorporated residual connections and a Dice-based loss function to improve volumetric segmentation Milletarì, Navab, and Ahmadi (2016). The nnU-Net framework later systematized this approach by automating architecture and training hyperparameters, establishing a robust and highly competitive baseline that remains a standard benchmark Isensee, Jaeger, Kohl, Petersen, and Maier-Hein (2020). These architectures share a core design principle: an encoder that progressively downsamples the input to capture semantic context, paired with a decoder that attempts to recover spatial detail.

A fundamental limitation of this successive downsampling strategy is the inherent trade-off between capturing expansive contextual information and preserving fine-grained spatial detail. Each pooling or strided convolution operation in the encoder irreversibly discards high-frequency information, geometrically reducing the spatial resolution of feature maps Badrinarayanan, Kendall, and Cipolla (2015). For the segmentation of sub-centimeter lesions, which may occupy only a handful of voxels, this compression creates a critical bottleneck. The spatial cues necessary to delineate subtle boundaries are attenuated or lost entirely in deeper encoder stages, before the decoder can attempt their recovery Chen, Papandreou, Kokkinos, Murphy, and Yuille (2016). Consequently, while these models excel at segmenting larger structures, their performance degrades significantly on small or low-contrast targets, as the decoder lacks the requisite high-resolution information to accurately reconstruct them Sunandini, Sivanpillai, Sowmya, and Variyar (2023).

To mitigate this resolution-context trade-off, numerous multi-scale and high-resolution aggregation techniques have been proposed. Atrous Spatial Pyramid Pooling (ASPP) and its variants employ parallel convolutions with different dilation rates to capture multi-scale context at a single feature resolution, avoiding further downsampling Qiu, Liu, Chen, Zhang, Zhu, and Xu (2023). Feature Pyramid Networks (FPN) construct a multi-scale feature pyramid by merging semantically strong, low-resolution features with high-resolution, semantically weak features from earlier encoder stages Lin et al. (2016). Architectural variants like U-Net++ introduced dense nested skip pathways to facilitate feature aggregation across multiple depths, aiming to bridge the semantic gap between encoder and decoder features Zhou, Siddiquee, Tajbakhsh, and Liang (2018). A more radical departure is seen in High-Resolution Net (HRNet), which maintains high-resolution representations throughout the network by connecting multi-resolution streams in parallel, thereby avoiding the typical information loss from downsampling.

However, these advanced techniques possess notable shortcomings for the specific challenge of tiny lesion segmentation. Methods like ASPP and FPN primarily operate on features that have already undergone significant downsampling; they aggregate context from a compressed representation but cannot regenerate the original high-frequency details that were discarded Lu, Sun, Yang, Song, Jiang, and Liu (2024). While HRNet preserves resolution, its design of maintaining multiple parallel streams at full and reduced resolutions leads to substantial computational and memory costs in 3D, limiting its practical application to high-resolution medical volumes. Techniques employing atrous convolutions uniformly across 3D volumes can suffer from gridding artifacts and may fail to capture truly long-range dependencies efficiently Chu, Tian, Zhang, Wang, and Shen (2021). Therefore, a persistent gap remains: the need for a mechanism that proactively preserves fine spatial detail throughout the encoding process, rather than attempting to recover it after the fact, which is particularly critical for the reliable segmentation of minuscule anatomical structures.

2.2. Techniques for High-Resolution and Contextual Modeling

A primary strategy to mitigate the loss of spatial detail in encoder-decoder networks involves maintaining high-resolution feature representations throughout the network's depth. The High-Resolution Network (HRNet) Yu, Xiao, Gao, Yuan, Zhang, Sang, and Wang (2021) pioneered this concept for 2D vision tasks by keeping a high-resolution stream in parallel with lower-resolution ones and repeatedly fusing information across scales. This design has been adapted to 3D medical segmentation to preserve fine-grained details for small structures. In contrast to methods like U-Net Ronneberger et al. (2015) that progressively discard resolution, these architectures aim to prevent the initial information loss. However, a direct 3D extension of HRNet's multi-branch fusion leads to prohibitive memory consumption due to the cubic growth of feature volumes, making it challenging to apply to high-resolution 3D medical images. Consequently, subsequent works have explored more efficient variants, such as maintaining only a single high-resolution residual stream with lightweight convolutions to reduce computational overhead while still preserving critical spatial cues Li, Zhang, Xiao, Zhang, and Bhanu (2022).

Transformer-based models have emerged as a powerful alternative for capturing long-range contextual dependencies, which are crucial for distinguishing small lesions from similar background tissue. Vision Transformers (ViTs) Dosovitskiy, Beyer, Kolesnikov, Weissenborn, Zhai, Unterthiner, Dehghani, Minderer, Heigold, and Gelly (2020) and their adaptations for medical imaging, such as UNETR and Swin UNETR Hatamizadeh, Nath, Tang, Yang, Roth, and Xu (2022), replace convolutional operations with self-attention mechanisms, enabling global receptive fields even at deep network stages. These models have demonstrated superior performance in segmenting structures with complex anatomical contexts. However, the standard self-attention mechanism scales quadratically with the number of tokens (voxels), making it computationally intractable for full-resolution 3D volumes Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, and Polosukhin (2017). While window-based approaches like Swin Transformers Hatamizadeh et al. (2022) offer linear complexity, they trade off global context for local window attention, which can be suboptimal for capturing evidence of tiny, dispersed lesions that require integrating information across large distances in the image volume.

To balance contextual modeling efficiency with the constraints of 3D data, several efficient alternatives have been developed. Atrous Spatial Pyramid Pooling (ASPP) uses dilated convolutions with different rates to capture multi-scale context without further downsampling, and has been effectively implemented in 3D segmentation backbones like nnU-Net Isensee, Petersen, Klein, Zimmerer, Jaeger, Kohl, Wasserthal, Koehler,

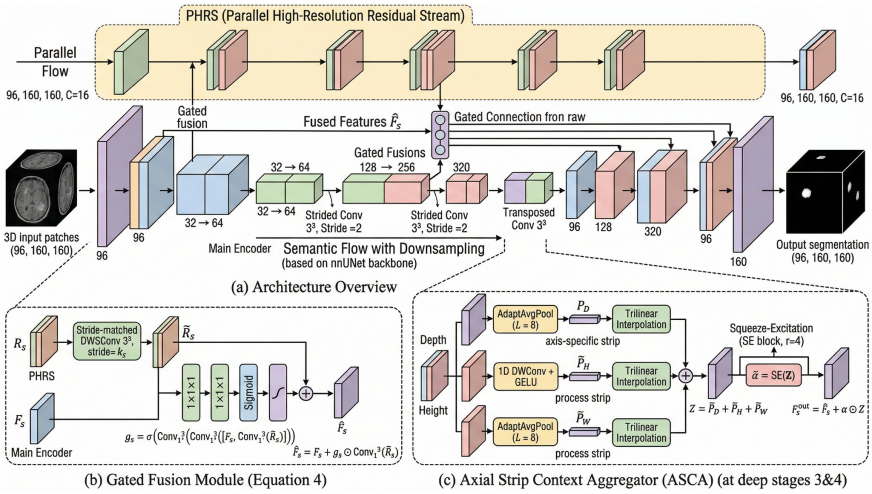

Figure 1: Overall architecture of HRFuse-Net. The framework integrates a Parallel High-Resolution Residual Stream (PHRS) and Axial Strip Context Aggregator (ASCA) modules into an nnUNet backbone, showing the data flow from input through the encoder-decoder pipeline to the final segmentation output.

Norajitra, and Wirkert (2018). However, uniformly applying large dilation rates across 3D volumes can lead to gridding artifacts and a loss of local coherence. Strip pooling Hong, Zhu, Liu, Ren, Shi, Lu, Yang, Deng, Qian, and Tan (2024) and axial attention decompose the global context aggregation into one-dimensional operations along each spatial axis. For instance, 3D axial attention applies self-attention independently along the height, width, and depth dimensions, achieving a receptive field that is global along each axis while maintaining linear complexity Al-Shabi, Shak, and Tan (2020). Similarly, methods employing lightweight depth-wise convolutions on pooled strip features provide an efficient mechanism for long-range dependency modeling Zhang, Zhu, Liu, and Liu (2020). While these strategies reduce computational cost compared to full self-attention, they often operate on features that have already undergone significant downsampling, thus struggling to recover the high-frequency details already lost in earlier encoder stages. Auxiliary techniques like self-distillation Zhang, Bao, and Ma (2021) and feature alignment Wang, Sun, and Zhao (2021) attempt to transfer detailed information from shallower to deeper layers, but they introduce additional training complexity and do not address the root cause of detail degradation in the primary forward path. Collectively, existing methods present a trade-off: high-resolution networks are memory-intensive in 3D, transformers are computationally prohibitive at full resolution, and efficient context modules often lack a mechanism for proactive detail preservation from the input stage onward.

3. Method

3.1. Architecture Overview

HRFuse-Net is a novel encoder-decoder architecture designed to address the fundamental challenge of preserving fine spatial detail for the segmentation of sub-centimeter brain metastases in 3D MRI. The framework is built upon the robust and well-established nnUNet backbone Isensee et al. (2018), which provides a systematic foundation for volumetric medical image segmentation. Our core innovation lies in augmenting this backbone with two synergistic components integrated into the encoder pathway: the Parallel High-Resolution Residual Stream (PHRS) and the Axial Strip Context Aggregator (ASCA). The PHRS proactively maintains a dedicated stream of high-resolution features throughout the encoding process, while the ASCA efficiently captures long-range, directional contextual information at deeper stages. This dual strategy ensures that both fine-grained boundary cues and semantically rich global context are preserved and effectively fused for the decoder's reconstruction task.

The network processes 3D input patches corresponding to the preprocessed dimensions of the MRI volumes. The encoder follows the standard nnUNet configuration with five hierarchical stages, progressively

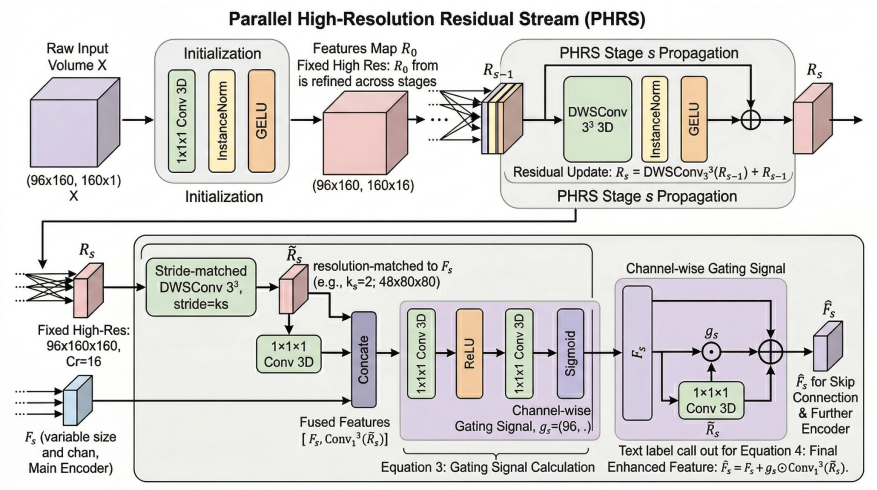

Figure 2: Detailed schematic of the Parallel High-Resolution Residual Stream (PHRS). The diagram illustrates the lightweight residual stream initialization, its fixed-resolution propagation, and the gated fusion mechanism for injecting high-resolution details into the main encoder pathway.

reducing spatial resolution to capture broader contextual information. Each stage transition is achieved via a strided $3 \times 3 \times 3$ convolution with a stride of 2. In parallel, the PHRS stream is initialized from the raw input and processed at the fixed original resolution without any downsampling. Before each encoder downsampling step, a gated fusion mechanism injects detail from this high-resolution stream into the corresponding encoder feature map, producing an enhanced feature representation $\hat{F}_s$ for each stage s. At the deepest encoder stages (stages 3 and 4), where semantic information is richest but spatial dimensions are manageable, the ASCA module is applied to $\hat{F}_s$ to aggregate axis-wise long-range context, yielding the final encoder output features F_s^{out}.

The decoder path mirrors the encoder's structure, utilizing transposed convolutions for upsampling. Crucially, it receives skip connections not from the original encoder features, but from the PHRS-enhanced features $\hat{F}_s$ at all corresponding stages. This ensures that the high-resolution spatial details, which have been preserved and reinforced by the PHRS, are directly available for the precise localization required during the segmentation mask reconstruction. The decoder blocks concatenate the upsampled features with the skip connections, followed by two $3 \times 3 \times 3$ convolutional blocks, each employing instance normalization Ulyanov, Vedaldi, and Lempitsky (2016) and the GELU activation function Lee (2023). The final layer is a $1 \times 1 \times 1$ convolution that projects the features to per-voxel class logits. The overall data flow, illustrated in Figure **??**, demonstrates how the PHRS and ASCA modules are seamlessly integrated into the encoder-decoder pipeline to form a cohesive, resolution-retentive segmentation framework.

3.2. Parallel High-Resolution Residual Stream

The Parallel High-Resolution Residual Stream (PHRS) constitutes the core mechanism of HRFuse-Net for proactively preserving fine spatial detail throughout the encoder pathway. Standard encoder-decoder architectures progressively downsample feature maps to capture semantic context, a process that irreversibly discards the sub-voxel boundary information critical for delineating brain metastases smaller than 0.02 cc Ronneberger et al. (2015). While multi-scale feature aggregation techniques like atrous spatial pyramid pooling attempt to recover context, they operate on already-degraded features and cannot regenerate lost high-frequency details. The PHRS addresses this fundamental limitation by maintaining a dedicated, fixed-resolution feature stream that operates in parallel to the main encoder, ensuring that high-resolution cues are never discarded but are instead continuously available for fusion.

The design of the PHRS is guided by the principle of minimal memory overhead to remain practical for 3D volumetric processing. The stream is initialized from the raw input volume X via a lightweight $1 \times 1 \times 1$ convolution with instance normalization and GELU activation, producing an initial feature map R_0 at the

original spatial resolution with a deliberately narrow channel width C_r. This low-dimensional representation is then propagated through all encoder stages without any spatial reduction. At each stage s, the stream is updated using a residual block built upon depth-wise separable $3 \times 3 \times 3$ convolutions, which efficiently refine features while preserving the fixed resolution. The update is defined as:

$$R_s = \mathrm{DWSConv}_{3^3}(R_{s-1}) + R_{s-1}, \tag{1}$$

where $\mathrm{DWSConv}_{3^3}$ denotes a depth-wise separable convolution with kernel size 3, padding 1, and stride 1, followed by instance normalization and GELU. This residual design ensures stable gradient flow and prevents feature degradation over the long propagation path.

The critical function of the PHRS is to inject preserved high-resolution details into the main encoder pathway before each downsampling step, thereby preventing irreversible information loss. This is achieved through a learned gated fusion mechanism. First, the high-resolution stream R_s is spatially aligned with the current encoder feature map F_s using a stride-matched projection. A depth-wise separable convolution with stride k_s—where k_s is the cumulative downsampling factor at stage s—projects R_s to the lower spatial dimensions:

$$\tilde{R}_s = \mathrm{DWSConv}_{3^3,\mathrm{stride}=k_s}(R_s). \tag{2}$$

A channel-wise gating signal g_s is then computed to adaptively control the fusion based on the content of both feature streams. The gate is generated by concatenating F_s with a $1 \times 1 \times 1$ convolution of $\tilde{R}_s$, processing the concatenated features through two successive $1 \times 1 \times 1$ convolutions, and applying a sigmoid activation σ:

$$g_s = \sigma\Big(\mathrm{Conv}_{1^3}\Big(\mathrm{Conv}_{1^3}([F_s,\, \mathrm{Conv}_{1^3}(\tilde{R}_s)])\Big)\Big). \tag{3}$$

The final enhanced encoder feature $\hat{F}_s$ is produced by a gated residual addition:

$$\hat{F}_s = F_s + g_s \odot \mathrm{Conv}_{1^3}(\tilde{R}_s). \tag{4}$$

This formulation allows the network to dynamically emphasize the most relevant high-resolution cues from the PHRS, complementing the semantically richer but coarser features in F_s.

By maintaining a constant, low channel count and employing efficient depth-wise separable convolutions, the PHRS introduces minimal computational and memory overhead—approximately 0.2M additional parameters—while fundamentally altering the information flow within the encoder. The fused features $\hat{F}_s$ are subsequently passed to the Axial Strip Context Aggregator at deeper stages and also serve as the skip connections for the decoder, ensuring that preserved fine details directly guide the final segmentation reconstruction. This design effectively decouples the need for spatial downsampling to gain context from the necessity of preserving resolution for accurate boundary delineation of small lesions.

3.3. Axial Strip Context Aggregator

The Axial Strip Context Aggregator (ASCA) is designed to efficiently capture long-range, directional contextual information at the deepest stages of the encoder, where semantic features are rich but spatial dimensions are sufficiently reduced. While the PHRS proactively preserves fine spatial detail, the complementary role of the ASCA is to enrich these features with global contextual cues that are essential for distinguishing small lesions from complex background tissue. This is achieved without resorting to computationally prohibitive full volumetric self-attention, which scales quadratically with voxel count Vaswani et al. (2017) and is impractical for high-resolution 3D MRI processing.

The module's design is based on the insight that long-range dependencies in volumetric medical images often exhibit strong directional patterns along anatomical axes. To model this efficiently, the ASCA employs a lightweight three-branch structure that processes context independently along the depth, height, and width axes. Each branch first collapses the input feature map along two spatial dimensions to form a compact strip representation, capturing a global summary along a single axis. Formally, given a PHRS-enhanced feature

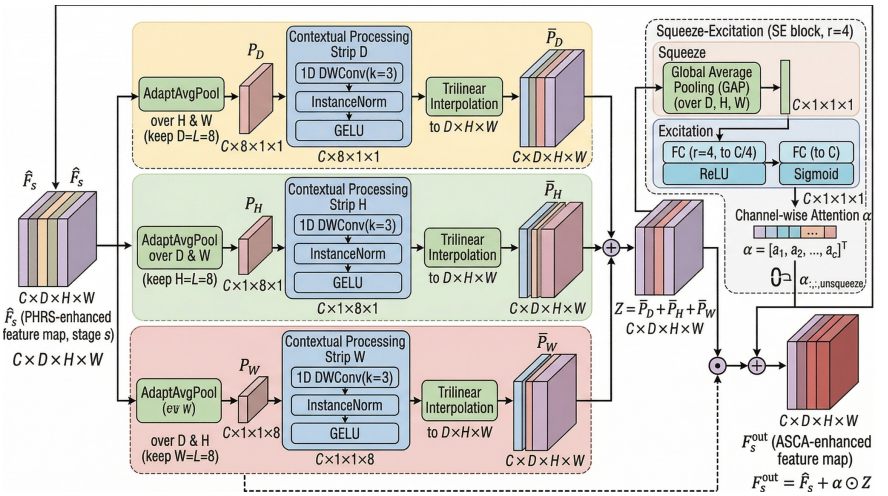

Figure 3: Structure of the Axial Strip Context Aggregator (ASCA). The figure depicts the three-branch axis-wise pooling, 1D depthwise convolution for context modeling, and the channel-wise recalibration via squeeze-excitation to aggregate long-range directional dependencies.

map $\hat{F}_s \in \mathbb{R}^{B\times C\times D\times H\times W}$ at stage s, the three parallel branches apply adaptive average pooling to reduce one axis to a fixed, short context length:

$$P_D = \text{AdaptAvgPool}_{(L,H,W)}(\hat{F}_s), \quad P_H = \text{AdaptAvgPool}_{(D,L,W)}(\hat{F}_s), \quad P_W = \text{AdaptAvgPool}_{(D,H,L)}(\hat{F}_s). \tag{5}$$

This pooling operation drastically reduces the computational footprint while preserving a condensed representation of axis-wise context. A 1D depthwise convolution with kernel size 3 is then applied separately along the pooled axis of each tensor to capture ordered contextual relationships within the strip, followed by instance normalization and a GELU activation:

$$\tilde{P}_d = \text{DWConv1D}_d(P_d), \quad d \in \{D, H, W\}. \tag{6}$$

Each processed strip is subsequently upsampled back to the original spatial dimensions (D, H, W) via trilinear interpolation, yielding axis-aware context maps $\bar{P}_D$, $\bar{P}_H$, and $\bar{P}_W$.

The three interpolated context maps are aggregated through summation to produce a comprehensive contextual feature $Z = \bar{P}_D + \bar{P}_H + \bar{P}_W$. To dynamically recalibrate the contribution of this aggregated context per channel, a squeeze-excitation block Hu, Shen, Albanie, Sun, and Wu (2017) is employed. This block first squeezes global spatial information from Z via global average pooling to produce a channel descriptor, which is then transformed through two fully connected layers with a reduction ratio r and a ReLU activation, followed by a sigmoid function to generate channel-wise attention weights $\alpha \in \mathbb{R}^{B\times C}$. The final output of the ASCA module is computed as a gated residual addition:

$$F_s^{\text{out}} = \hat{F}_s + \alpha_{:,:,:,\text{unsqueeze}} \odot Z. \tag{7}$$

This design allows the network to selectively emphasize the most informative contextual signals from each axis. The ASCA is applied specifically to the PHRS-enhanced features $\hat{F}_s$ at encoder stages 3 and 4, where the spatial dimensions are compact enough for efficient strip-based computation. Its computational cost scales linearly with the spatial dimensions due to the independent axis-wise processing, offering a significant efficiency advantage over quadratic-cost self-attention mechanisms Dosovitskiy et al. (2020) while still providing the long-range contextual modeling necessary for accurate small lesion segmentation.

4. Experiments

4.1. Datasets and Implementation Details

We evaluate our proposed HRFuse-Net on the publicly available *LMD-BM* dataset, a longitudinal MRI dataset of brain metastases with tumor segmentations, clinical and radiomic data Ramakrishnan, Jekel, Chadha, Janas, Moy, Maleki, Sala, Kaur, Petersen, and Merkaj (2024). This dataset comprises 44 patient cases with multi-sequence MRI scans, including FLAIR, T1-weighted, post-contrast T1-weighted (T1c), and T2-weighted sequences. The segmentation task involves delineating three key tumor subregions: necrotic core, peritumoral edema, and enhancing tumor, resulting in a four-class segmentation problem. All images were acquired from 1.5T and 3T scanners and underwent a standardized preprocessing pipeline, including co-registration to the BraTS spatial framework and intensity normalization. The ground truth annotations were generated through a combination of automated segmentation methods followed by meticulous manual correction and final review by an expert neuroradiologist with over a decade of experience. Accurate segmentation of these subregions is clinically crucial for stereotactic radiosurgery planning and for monitoring treatment response during longitudinal follow-up. We adopt an 8:2 train-test split, resulting in 35 cases for training and 9 for testing, consistent with standard evaluation protocols for medical image segmentation.

We employ standard volumetric segmentation metrics to quantitatively assess performance, with a focus on boundary precision critical for small lesions. The primary metrics are the Dice Similarity Coefficient (DSC) and the 95th percentile Hausdorff Distance (HD95). The DSC measures volumetric overlap between the predicted segmentation P and the ground truth G:

$$\mathrm{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{8}$$

A higher DSC indicates better overall region agreement. For boundary accuracy, we compute the HD95, which represents the 95th percentile of the maximum surface distances between the two segmentations, making it robust to outliers. A lower HD95 signifies superior boundary delineation, with clinical guidelines often targeting values below 4mm for precise surgical or radiosurgical planning. We also report sensitivity and specificity to evaluate per-voxel classification performance, providing a comprehensive view of the model's ability to correctly identify both lesion and background voxels.

Our implementation is built upon the nnU-Net framework Niu, Zhang, Zhou, Feng, and Ling (2025) using its *3d_fullres* configuration. The proposed HRFuse-Net was trained for 1000 epochs with a batch size of 2 on an NVIDIA GeForce RTX 4090 GPU. We used stochastic gradient descent (SGD) with Nesterov momentum (0.99), a weight decay of 3×10^{-5}, and an initial learning rate of 0.01 governed by a polynomial decay scheduler. The loss function combines Dice and cross-entropy terms. Input patches of size $96\times160\times160$ were randomly sampled during training, applying the standard nnU-Net data augmentation pipeline including rotation, scaling, and elastic deformations. All preprocessing (e.g., resampling, normalization) and postprocessing steps strictly follow the nnU-Net pipeline to ensure a fair comparison. For non-nnU-Net baseline models, we adhere to the principle of using method-specific configurations for fairness. When available, we employ the original implementation or the authors' recommended architectural and hyperparameter settings; otherwise, the official default configuration is used. All baselines share the same nnU-Net preprocessing and postprocessing pipeline, train-test split, and are evaluated on identical hardware to isolate architectural contributions.

4.2. Ablation Studies

We conduct comprehensive ablation studies to validate the contribution of each architectural component and the design choices within HRFuse-Net. The experiments systematically evaluate the impact of removing core modules and compare alternative design variants against our selected configuration.

4.2.1. Component Removal Studies

To quantify the contribution of each proposed module, we evaluate the performance degradation when individual components are removed. Table 1 presents the results. The full HRFuse-Net configuration achieves a Dice score of 0.5101 and an HD95 of 20.2752 mm. Removing the Parallel High-Resolution Residual Stream (PHR) results in the most severe performance drop, reducing the Dice score to 0.4362 and increasing HD95 to 30.6883 mm. This substantial decline emphasizes the critical role of the PHR in preserving fine spatial

Table 1
Component ablation results. Removing any proposed module leads to significant performance degradation.

Configuration	Removed Component	Dice Score	HD95 Score
Full Model	-	**0.5101**	**20.2752**
w/o PHR	PHR	0.4362	30.6883
w/o ASCA	ASCA	0.4687	31.6258
w/o GF	GF	0.4405	28.3944

Table 2
Ablation study on the context length L for the ASCA module.

Variant	Dice Score	HD95 Score
Proposed Method ($L = 4$)	**0.5101**	**20.2752**
$L = 8$	0.4762	24.0340
$L = 12$	0.4469	27.4002
$L = 16$	0.4709	26.1989
Adaptive Pooling	0.4817	23.5962

detail throughout the encoder, which is essential for delineating sub-centimeter metastases. Disabling the Axial Strip Context Aggregator (ASCA) module yields a Dice score of 0.4687 and an HD95 of 31.6258 mm, confirming that the long-range, axis-wise contextual information it provides is vital for accurate semantic segmentation. Finally, removing the Gated Fusion (GF) mechanism and replacing it with simple addition leads to a Dice score of 0.4405 and an HD95 of 28.3944 mm. This result validates the importance of the learned gating mechanism in selectively integrating high-resolution features from the PHR stream, preventing the injection of irrelevant or noisy details that could hinder the learning process.

4.2.2. ASCA Context Length Analysis

The Axial Strip Context Aggregator captures long-range dependencies by pooling features along each spatial axis to a fixed context length. We investigate the sensitivity of performance to this key hyperparameter. Table 2 compares variants with different context lengths, including an adaptive pooling strategy. The proposed method, which uses a context length of 4 ($L = 4$), achieves the best performance. Increasing the context length to 8, 12, or 16 leads to a consistent decrease in Dice score and an increase in HD95. This suggests that over-aggregating context along the axes may dilute locally relevant features for small lesions. The adaptive pooling variant, which does not fix the context length, performs better than the fixed $L \geq 8$ variants but is still outperformed by $L = 4$. The optimal performance at $L = 4$ indicates that for the task of segmenting small brain metastases, a compact but structured representation of axial context is more effective than very long-range or fully adaptive aggregation, likely because it provides sufficient contextual guidance without sacrificing the local specificity needed for tiny targets.

4.2.3. PHR Fusion Strategy Comparison

The mechanism for fusing information from the Parallel High-Resolution Residual Stream into the main encoder pathway is a crucial design choice. We evaluate several alternative fusion strategies against our gated residual approach. The results are shown in Table 3. The proposed gated fusion strategy significantly outperforms all alternatives. Simple addition or concatenation of the projected PHR features leads to Dice scores of 0.3957 and 0.4546, respectively. A squeeze-and-excitation (SE) attention mechanism for fusion performs only marginally better than addition, with a Dice score of 0.4103. Completely disabling the fusion (labeled *none*) results in the worst performance, with a Dice score of 0.3618. This comprehensive comparison demonstrates that a naive integration of high-resolution details can be detrimental, likely because it introduces unrefined, low-level features that conflict with the semantically richer features in the deeper encoder stages. The success of the gated fusion mechanism lies in its ability to learn a spatially and channel-wise adaptive weighting, allowing the network to selectively emphasize only the most relevant high-resolution cues needed to refine the feature maps at each stage.

Table 3
Comparison of different fusion strategies for integrating the PHR stream.

Fusion Strategy	Dice Score	HD95 Score
Proposed (Gated)	**0.5101**	**20.2752**
Addition	0.3957	31.6225
Concatenation	0.4546	27.3954
SE Attention	0.4103	29.3252
None	0.3618	38.2462

Table 4
State-of-the-art comparison on the LMD-BM dataset.

Method	Dice Score	HD95
nnUNet Niu et al. (2025)	0.4619	32.5388
ThreeDUXNet Lee, Bao, Huo, and Landman (2022)	0.4371	29.1306
UNetPlusPlus Zhou et al. (2018)	0.4321	35.0007
nnFormer Zhou, Guo, Zhang, Yu, Wang, and Yu (2021)	0.4320	24.5692
STUNet	0.4056	28.3254
UNETR Hatamizadeh, Yang, Roth, and Xu (2021)	0.3963	35.2232
UMamba Liu, Yang, Zhou, Xi, Yu, Yu, Liang, Shi, Zhang, and Zheng (2024)	0.3610	31.5977
HRFuse-Net (Ours)	**0.5069**	**25.4869**

4.3. State-of-the-Art Comparisons

We comprehensively evaluate HRFuse-Net against state-of-the-art segmentation models on the LMD-BM dataset. The quantitative comparison, presented in Table 4, demonstrates that our method achieves superior performance on the challenging task of small brain metastasis segmentation.

HRFuse-Net attains the highest Dice score of 0.5069, representing a 9.75% improvement over the strongest baseline, nnUNet (0.4619). This significant gain in volumetric overlap is complemented by a substantial 21.67% reduction in the 95th percentile Hausdorff Distance (HD95), from 32.5388 mm to 25.4869 mm, indicating superior boundary delineation. Notably, while nnFormer achieves the best HD95 among baselines (24.5692), its Dice score is considerably lower (0.4320). HRFuse-Net strikes a more favorable balance, offering the best Dice while maintaining a competitive HD95, second only to nnFormer. This performance advantage stems directly from our architectural design: the Parallel High-Resolution Residual Stream (PHRS) proactively preserves fine spatial details that are critical for identifying sub-centimeter lesions, while the Axial Strip Context Aggregator (ASCA) enriches features with long-range anatomical context without the quadratic cost of full self-attention. This combination effectively addresses the core limitation of standard encoder-decoder networks, where successive downsampling irreversibly discards the sub-voxel cues needed for small target segmentation.

The distribution of per-case Dice scores, visualized in Figure 4, reveals that HRFuse-Net achieves a higher mean and exhibits more consistent performance across the patient cohort compared to baseline methods. The reduced variance suggests improved robustness to the heterogeneous appearance and size of brain metastases. Qualitative analysis in Figure 5 corroborates these findings, showing that our method produces segmentation masks with more accurate boundaries and fewer false positives, particularly for small and low-contrast lesions. The clinical relevance of this improvement is highlighted by the cumulative distribution of boundary errors in Figure 6; a lower HD95 distribution is crucial for treatment planning applications like stereotactic radiosurgery, where precise spatial localization is paramount. Figure 7 demonstrates that the performance advantage of HRFuse-Net is most pronounced for the smallest lesions ($<$0.006 cc), validating that the resolution-preserving mechanism of the PHRS is specifically beneficial for this challenging subgroup.

The computational profile of HRFuse-Net represents a practical trade-off. With 19.81 million parameters, it is more compact than nnUNet (88.22M) and UNETR (95.61M), and comparable to SegResNet (18.80M). Its FLOPs (1833.21G) and inference time (126.7 ms) are higher than several efficient transformers like nnFormer and SwinUNETR, but remain within a feasible range for clinical deployment. This moderate

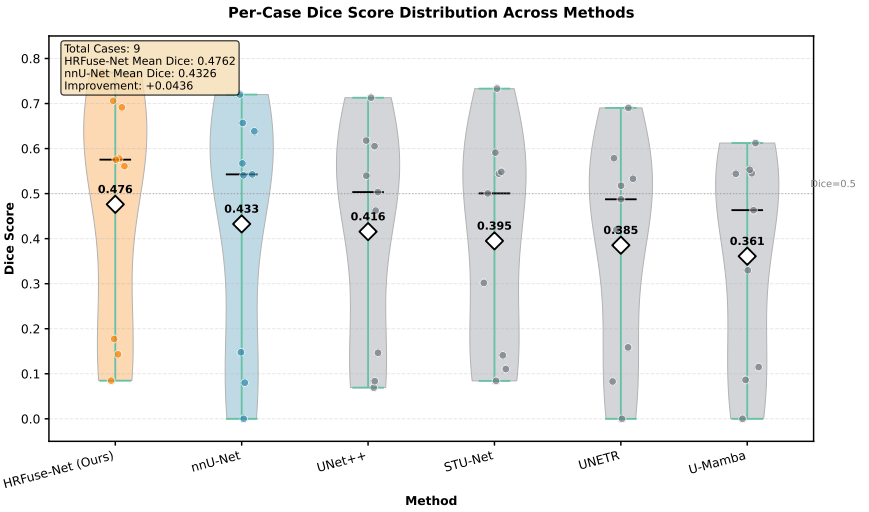

Figure 4: Violin plots with swarm overlay showing the distribution of per-case Dice scores across all methods. The width of each violin represents the probability density of scores. Individual cases are shown as points, and diamond markers indicate the mean value for each method. HRFuse-Net achieves the highest mean Dice (0.5069) and demonstrates more consistent performance across cases compared to baselines.

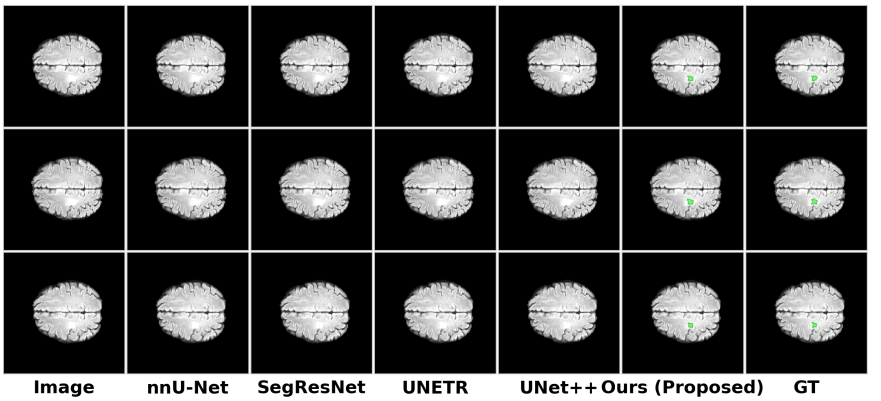

Figure 5: Qualitative comparison of segmentation results. Columns show the input image, predictions from baseline methods (nnU-Net, SegResNet, UNETR, UNet++), our proposed method, and the ground truth. Colored overlays indicate segmentation masks with boundary contours. Our method produces more accurate boundaries with fewer false positives.

increase in computational cost is justified by the substantial gains in segmentation accuracy for small lesions, a trade-off that is often necessary for tackling this specific and clinically significant challenge.

5. Discussion

5.1. Interpretation of Results and Implications

The experimental results demonstrate that HRFuse-Net achieves state-of-the-art performance on the challenging task of segmenting small brain metastases, as evidenced by its superior Dice score of 0.5069 and a substantially improved HD95 of 25.4869 mm compared to all evaluated baselines. The 9.75% improvement in Dice and 21.67% reduction in HD95 over the strong nnUNet baseline are particularly significant given the difficulty of the dataset, which contains numerous lesions smaller than 0.02 cc. The qualitative analysis in Figure 5 visually confirms that our method produces more accurate boundaries and fewer false positives, especially for subtle, low-contrast lesions. The distribution of per-case Dice scores, shown in Figure 4,

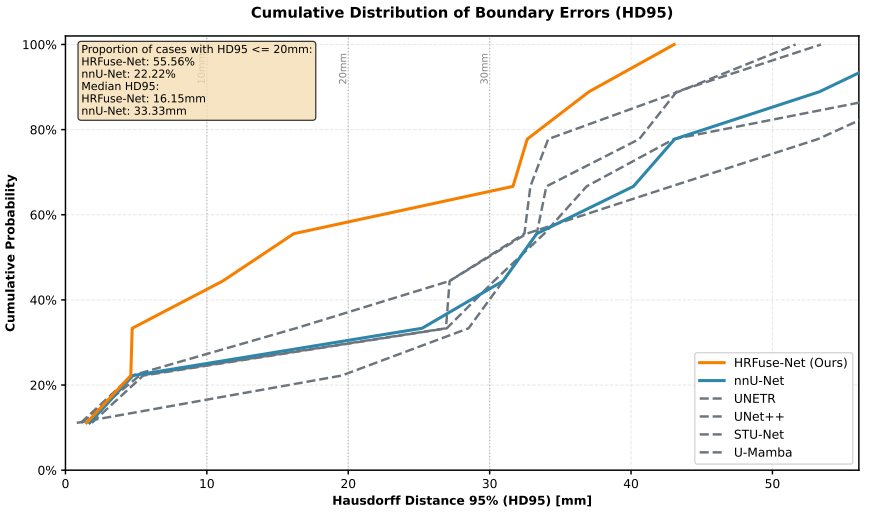

Figure 6: Cumulative Distribution of Boundary Errors (HD95)

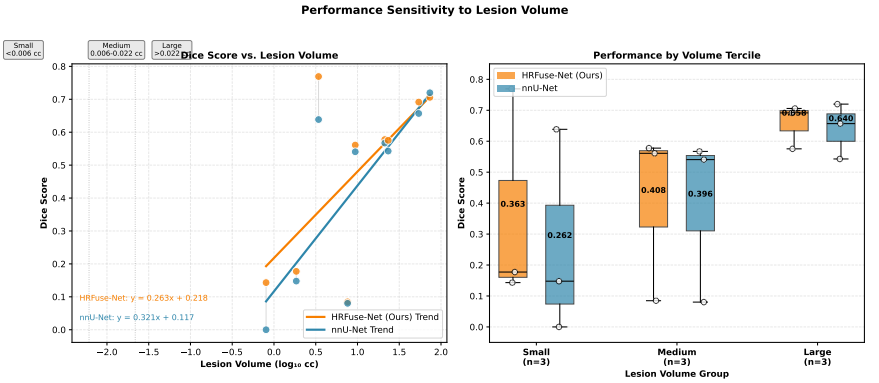

Figure 7: Left: Scatter plot with regression lines showing Dice score as a function of lesion volume (log scale). HRFuse-Net shows a steeper positive slope, indicating better performance improvement with decreasing lesion size. Right: Box plots grouped by volume terciles demonstrate that HRFuse-Net's advantage is most pronounced for small lesions (<0.006 cc), with a mean Dice improvement of +0.126 over nnU-Net in this challenging subgroup.

indicates that HRFuse-Net achieves a higher mean and exhibits more consistent performance across cases, reducing the variance in segmentation quality. This robustness is a critical requirement for clinical translation, where reliability is paramount.

The core implication of these findings is that the fundamental architectural principle of hierarchical resolution-retentive feature encoding is highly effective for small-object segmentation in 3D medical images. The success of the Parallel High-Resolution Residual Stream (PHRS) validates our central hypothesis: proactively preserving fine spatial detail from the input stage onward is more effective than attempting to recover it after it has been lost through downsampling. By maintaining a dedicated high-resolution pathway and fusing its features into every encoder stage via a learned gated mechanism, the network prevents the irreversible compression of sub-voxel boundary cues. This design directly addresses the primary limitation of standard encoder-decoder networks and late-stage multi-scale aggregation methods like ASPP. The significant performance drop observed when removing the PHRS in ablation studies (Dice: 0.4362) emphasizes its indispensable role.

The complementary role of the Axial Strip Context Aggregator (ASCA) is evidenced by its contribution to capturing long-range, anatomically coherent contextual information. While the PHRS preserves local

detail, the ASCA efficiently models global relationships along the principal anatomical axes, which is crucial for distinguishing metastases from similarly appearing background tissue and for understanding the spatial layout of complex, multi-focal disease. The ablation study confirms that removing the ASCA also leads to a notable performance degradation (Dice: 0.4687). The synergy between these two components—local detail preservation and efficient global context aggregation—forms the foundation of HRFuse-Net's effectiveness, offering a balanced solution to the classic trade-off between receptive field size and spatial resolution.

5.2. Comparison with State-of-the-Art Methods

HRFuse-Net outperformed a comprehensive set of contemporary architectures, including transformer-based models (UNETR, nnFormer Dora, Agrawal, Panda, and Abraham (2017)), efficient convolutional networks (nnUNet Isensee et al. (2020)), and recent hybrids (UMamba Liu et al. (2024)). Notably, it surpassed nnFormer, which achieved the best HD95 among baselines (24.5692 mm), by a significant margin in Dice score (+0.0749). This suggests that while pure transformer models can capture excellent long-range context, their inability to process features at full 3D resolution due to quadratic complexity limits their ability to delineate the finest details of tiny lesions. Our ASCA module provides a computationally linear alternative for context modeling that operates on features already enriched by the PHRS, yielding a more favorable accuracy-efficiency profile for this specific task.

The method also shows a clear advantage over models like HRNet Sun, Xiao, Liu, and Wang (2019), which maintain multiple resolution streams in parallel. A direct 3D extension of HRNet would be prohibitively memory-intensive. HRFuse-Net's innovation lies in maintaining only a single, lightweight high-resolution stream (with a fixed channel width of $C_r = 16$) that is selectively fused into the main encoder, achieving the benefit of multi-resolution representation with minimal overhead. The cumulative distribution of boundary errors in Figure 6 shows that HRFuse-Net achieves lower HD95 values for a greater proportion of cases compared to other methods, indicating superior boundary localization accuracy that is consistent with the visual results.

The analysis of performance sensitivity to lesion volume, presented in Figure 7, provides the most compelling evidence for the method's targeted efficacy. The steeper positive slope of the regression line for HRFuse-Net indicates that its performance advantage increases as lesion size decreases. For the smallest lesions ($<$0.006 cc), our method achieved a mean Dice improvement of +0.126 over nnUNet. This subgroup analysis confirms that the architectural innovations are specifically beneficial for the most challenging cases that are often missed or poorly segmented by conventional networks, directly addressing the clinical need for reliable detection and delineation of early-stage or residual disease.

5.3. Limitations and Future Work

Despite its strong performance, this study has several limitations. First, the evaluation was conducted on a single, publicly available dataset (LMD-BM) with 44 cases. While the dataset is well-annotated and represents a realistic clinical challenge, further validation on larger, multi-institutional cohorts is necessary to establish generalizability across different MRI scanners, protocols, and patient populations. The performance on very small lesions, though improved, remains imperfect (Dice 0.51), highlighting that segmenting sub-centimeter metastases from MRI alone remains an unsolved problem. Incorporating complementary imaging modalities or clinical metadata could provide additional cues.

Second, the computational efficiency of HRFuse-Net presents a trade-off. With 19.81M parameters and 1833.21 GFLOPs, it is more computationally demanding than some highly efficient baselines like MedNeXt (5.98M params) and UNETR (284.18 GFLOPs), though it remains more efficient than the original nnUNet backbone (4350.66 GFLOPs). The inference time of 126.7 ms per patch is acceptable for offline analysis but may require optimization for real-time clinical workflows. Future work could explore more aggressive model compression, knowledge distillation, or the use of dynamic neural networks where the PHRS stream is activated only for patches suspected of containing small lesions.

The design of the ASCA module, while efficient, captures context along axes independently and may not fully model more complex, non-axis-aligned anatomical relationships. Exploring hybrid attention mechanisms that combine the efficiency of strip pooling with sparse or window-based global attention could be a fruitful direction. The current gated fusion mechanism operates at the channel level; investigating spatial-adaptive

gating that modulates fusion based on local image content could further refine the integration of high-resolution details.

Finally, the clinical utility of the segmentations should be evaluated in downstream tasks, such as radiomics feature extraction for prognosis prediction or volumetric measurement for treatment response assessment. A prospective study correlating the accuracy of automated segmentations with clinical outcomes would be the ultimate validation of the method's impact.

6. Conclusion

In this work, we presented HRFuse-Net, a novel 3D encoder-decoder architecture designed for the precise segmentation of very small brain metastases. By introducing a Parallel High-Resolution Residual Stream (PHRS) that preserves fine spatial detail throughout the encoding process and an Axial Strip Context Aggregator (ASCA) that captures long-range dependencies efficiently, the method effectively bridges the resolution-context trade-off inherent in standard networks. Comprehensive experiments on a public longitudinal MRI dataset demonstrate that HRFuse-Net sets a new state-of-the-art, with particular strength in segmenting the smallest and most challenging lesions. The architectural principles are general and could be adapted to other medical imaging tasks where the accurate delineation of fine structures is critical, such as segmenting small vessels, lung nodules, or focal cortical dysplasias. This work provides a promising direction for developing more reliable automated tools to assist in the early detection and management of metastatic brain disease.

Extensive evaluation on the publicly available LMD-BM dataset demonstrates the effectiveness of our approach. HRFuse-Net achieved state-of-the-art performance, with a Dice score of 0.5069 and an HD95 of 25.4869 mm, representing a 9.75% improvement in Dice and a 21.67% reduction in HD95 over the strong nnU-Net baseline. Ablation studies confirmed the critical contribution of each component, with the removal of the PHRS causing the most significant performance degradation, emphasizing the importance of proactive resolution preservation. The analysis revealed that HRFuse-Net's advantage is most pronounced for the smallest lesions ($<$0.006 cc), as shown in Figure 7, demonstrating its targeted efficacy for the most challenging cases. The model exhibited more consistent performance across cases, as evidenced by the distribution of per-case Dice scores in Figure 4.

In conclusion, HRFuse-Net successfully bridges the critical gap between high-resolution detail preservation and efficient long-range context modeling for 3D medical image segmentation. By ensuring that fine spatial cues are never lost but continuously guide the learning of semantic features, the architecture provides a principled solution to the small lesion segmentation problem. The design principles are general and could benefit other medical imaging tasks requiring precise delineation of fine structures. Future work will focus on validating the model's generalizability across larger, multi-center cohorts and exploring further optimizations for integration into clinical workflows.

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